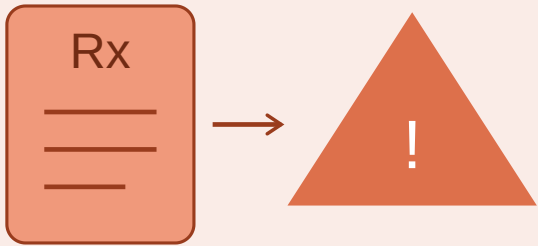


Explainable CNN Framework for Handwritten English Alphabet Recognition

1

Practical Problem

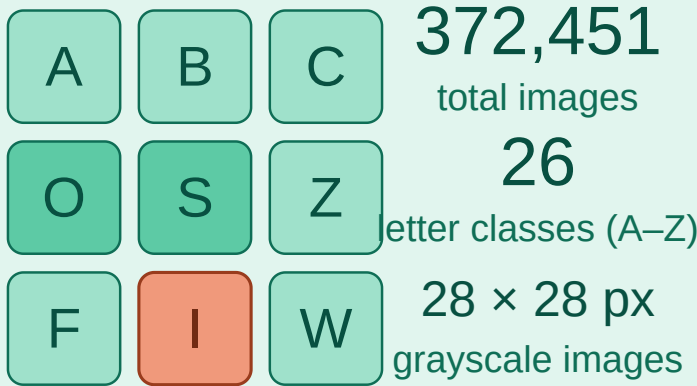


Handwritten Rx Patient harm risk

OCR · document digitisation
assistive technology
→ need for reliable letter recognition

2

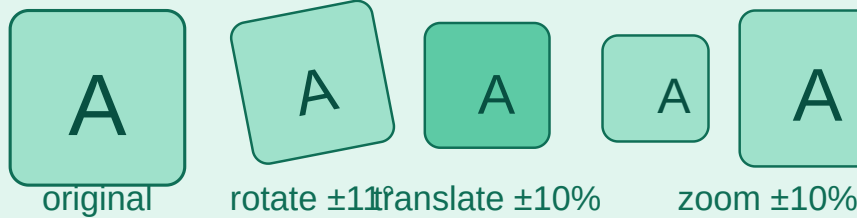
Dataset



Class imbalance: 51.6 : 1 (O = 57,825 · I = 1,120)
■ majority ■ minority

3

Preprocessing & Augmentation



Normalise pixels:

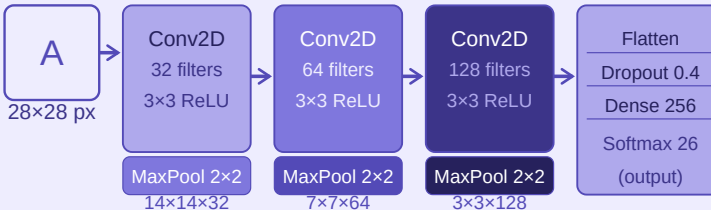
0–255 (raw) → 0–1 (normalised)

Class weights (balanced):

0 · S (majority)
F · I (minority — up-weighted)

4

Custom CNN (Baseline)

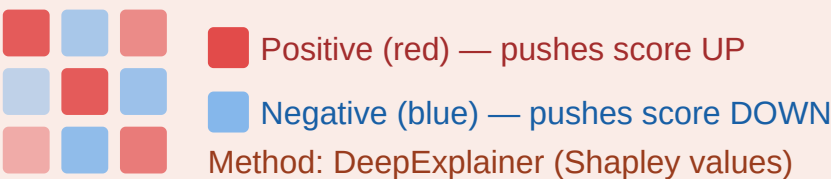


394,522 parameters · < 5 MB model file

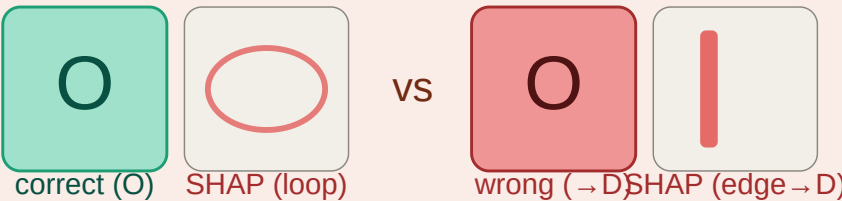
Test accuracy: 99.08%
Macro F1: 98.80% · AUC: 1.000
All 26 letter classes F1 > 0.96

8

SHAP Explainability

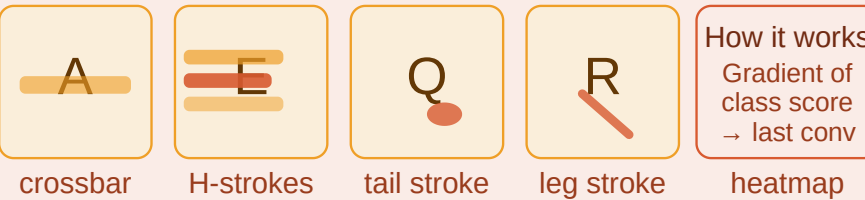
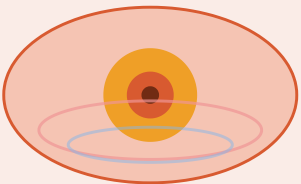


Correct O vs wrong O → D prediction



7

Grad-CAM Explainability



Model focuses on discriminative stroke regions — not background

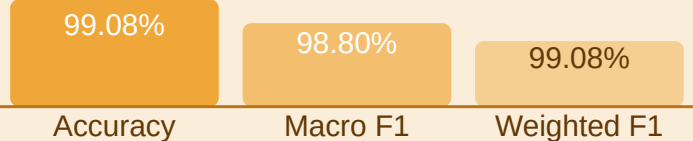
6

Evaluation

99.08%

Test accuracy (Custom CNN)

AUC
1.000

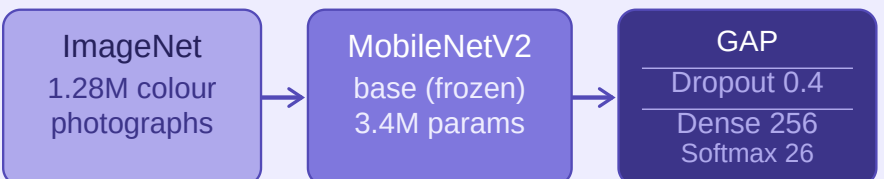


Custom CNN: 99.08% ★ vs MobileNetV2: 19.59%

Minority class recall — class weighting effective
F (1,163 imgs): 0.994 I (1,120 imgs): 0.976

5

MobileNetV2 Transfer Learning



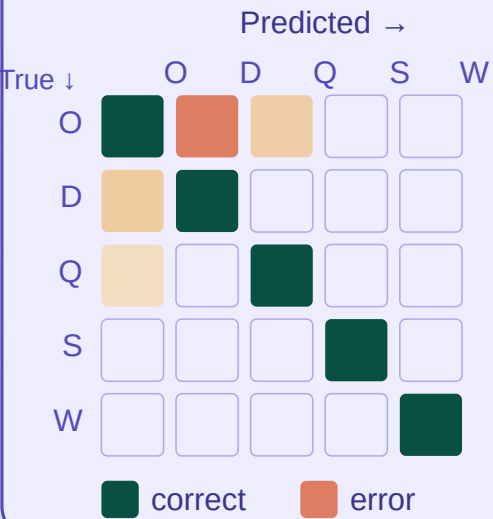
Stage 1: freeze base · train head · 5 epochs · lr=1e-3

Stage 2: unfreeze top 30 layers · fine-tune · lr=1e-5

Domain mismatch: colour photos ≠ grayscale letters
Result: 19.59% accuracy (transfer learning fails here)

9

Confusion Matrix & Error Analysis



Top confused pairs

O → D 64 cases
D → O 32 cases
O → Q 25 cases
W ↔ N 13 cases
N → H 8 cases

All caused by visual stroke similarity
not data scarcity — strong diagonal across all 26 classes

10

Practical Outcome

★
99.08%
test accuracy · 26 classes
F1: 98.80%
AUC: 1.000

CNN outperforms MobileNetV2
99.08% vs 19.59%

Key achievements

394,522 params · <5MB Mobile / edge deployable
Grad-CAM + SHAP verified Reproducible pipeline

Applications

OCR · Medical Rx · Document digitisation
Assistive technology · Character recognition

Domain mismatch prevents transfer learning benefit